

Course Code: Ethical Hacking Laboratory

Software Required: Kali Linux (Latest Version)

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Experiment 1

Basic File and User Management Commands

Aim

To execute and understand the usage of **rm**, **users**, and **tree** commands in Kali Linux.

Procedure

1. Open Kali Linux Terminal.
2. Create a sample file.
3. Execute the rm command.
4. Execute the users command.
5. Execute the tree command.

Commands:

Touch test.txt

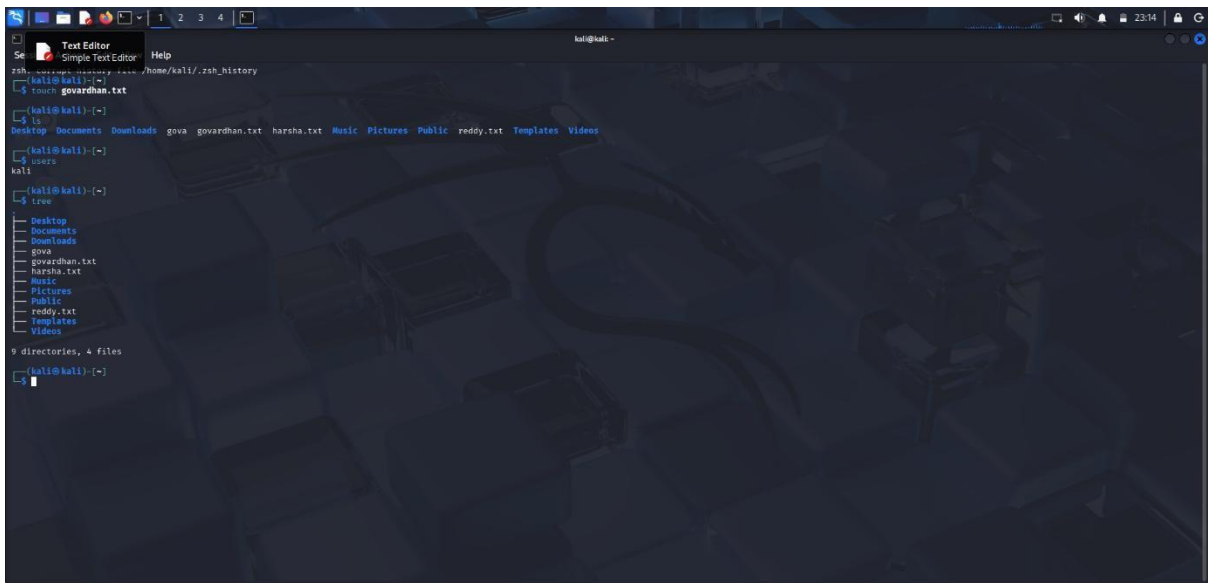
Ls

Rm test.txt

User

Tree

Output



```
kali@kali: ~  
$ rm govardhan.txt  
$ users  
kali  
$ tree  
.  
├── Desktop  
├── Documents  
├── Downloads  
├── gova  
├── govardhan.txt  
├── harsha.txt  
├── Music  
├── Pictures  
├── Public  
├── reddy.txt  
├── Templates  
└── Videos  
9 directories, 4 files  
$
```

Result

Successfully executed rm, users, and tree commands in Kali Linux.

Experiment 2

File Viewing Commands

Aim

To execute and understand the usage of **less**, **more** and **vi** commands.

Procedure

1. Create a text file.
2. Open the file using less.
3. Open the file using more.
4. Edit the file using vi.

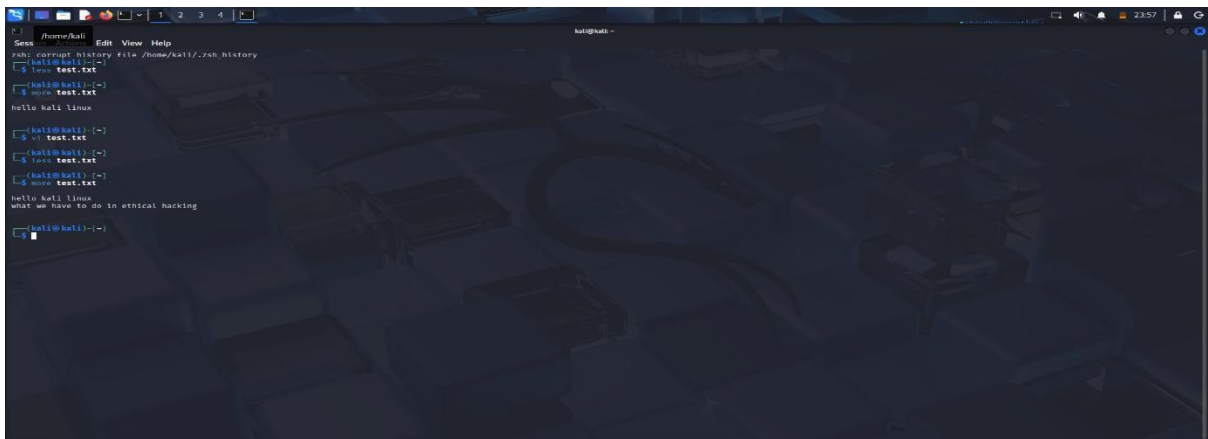
Commands:

less test.txt

more test.txt

vi test.txt

Output

A screenshot of a terminal window with a dark background and a cityscape wallpaper. The terminal shows a series of commands and their outputs. The user starts by running 'cat /dev/urandom | tr -dc 'a-z' | fold -w 60 | xargs -n 1 shuf | paste | shuf | sed 'N;N;\$!D;s/ / /g'> test.txt'. Then they run 'cat test.txt' which outputs 'hello kali linux'. Next, they run 'nano test.txt' which opens the nano text editor. They press 'v' to enter visual mode, then 'i' to enter insert mode, and type 'what we have to do in ethical hacking'. Finally, they press 'Ctrl+X' to save and exit, and the terminal shows 'hello kali linux' and 'what we have to do in ethical hacking' on separate lines.

```
kali@kali:~$ cat /dev/urandom | tr -dc 'a-z' | fold -w 60 | xargs -n 1 shuf | paste | shuf | sed 'N;N;$!D;s/ / /g'> test.txt
kali@kali:~$ cat test.txt
hello kali linux
kali@kali:~$ nano test.txt
^V^I
hello kali linux
what we have to do in ethical hacking
^X
hello kali linux
what we have to do in ethical hacking
kali@kali:~$
```

Result

Successfully executed file viewing and editing commands.

Experiment 3

Tree, Head and Tail Commands

Aim

To execute and understand the usage of tree, head, and tail commands.

Procedure

1. Create a text file containing multiple lines.
2. Execute the head command.
3. Execute the tail command.
4. Display directory structure using tree.

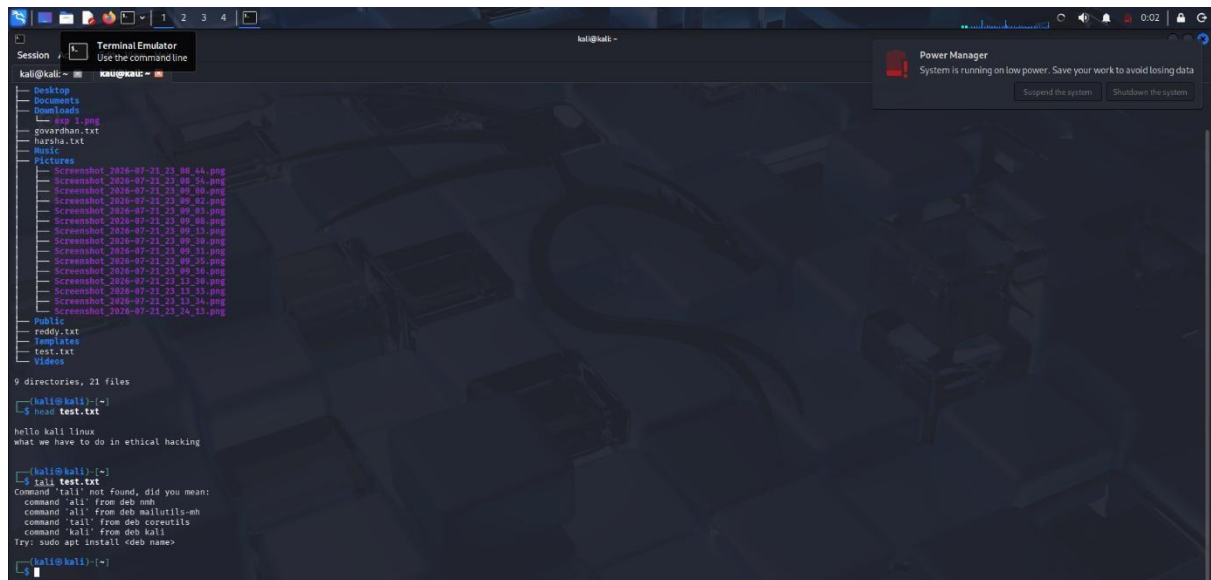
Commands:

Tree

Head test.txt

Tali test.txt

Output

A screenshot of a Kali Linux terminal window. The terminal shows a file tree listing directories like Desktop, Documents, Downloads, and files like Luv.jpg, govardhan.txt, and harsha.txt. It also lists a 'Pictures' directory with numerous screenshot files. The user runs the 'head test.txt' command, which outputs 'Hello kali linux' and 'what we have to do in ethical hacking'. Then, the user runs 'tail test.txt', which shows an error message: 'Command 'tail' not found, did you mean: command 'all' from deb mh, command 'all' from deb mailutils-mh, command 'tail' from deb coreutils, command 'kali' from deb kali. Try: sudo apt install <deb name>'.

```
kali@kali: ~  
Desktop  
Documents  
Downloads  
Luv.jpg  
govardhan.txt  
harsha.txt  
Music  
Pictures  
Screenshot_2020-07-21_23_08_44.png  
Screenshot_2020-07-21_23_08_54.png  
Screenshot_2020-07-21_23_09_00.png  
Screenshot_2020-07-21_23_09_02.png  
Screenshot_2020-07-21_23_09_03.png  
Screenshot_2020-07-21_23_09_08.png  
Screenshot_2020-07-21_23_09_13.png  
Screenshot_2020-07-21_23_09_30.png  
Screenshot_2020-07-21_23_09_31.png  
Screenshot_2020-07-21_23_09_35.png  
Screenshot_2020-07-21_23_09_36.png  
Screenshot_2020-07-21_23_10_26.png  
Screenshot_2020-07-21_23_11_35.png  
Screenshot_2020-07-21_23_13_34.png  
Screenshot_2020-07-21_23_14_13.png  
Public  
redoy.txt  
templates  
test.txt  
Videos  
9 directories, 21 files  
kali@kali:~$ head test.txt  
Hello kali linux  
what we have to do in ethical hacking  
kali@kali:~$ tail test.txt  
Command 'tail' not found, did you mean:  
command 'all' from deb mh  
command 'all' from deb mailutils-mh  
command 'tail' from deb coreutils  
command 'kali' from deb kali  
Try: sudo apt install <deb name>  
kali@kali:~$
```

Result

Successfully executed tree, head, and tail commands.

Experiment 4

Directory and File Display Commands

Aim

To execute ls, cat, and mkdir commands.

Procedure

1. Create a directory using mkdir.
2. List files using ls.
3. Create a text file.
4. Display file contents using cat.

Commands:

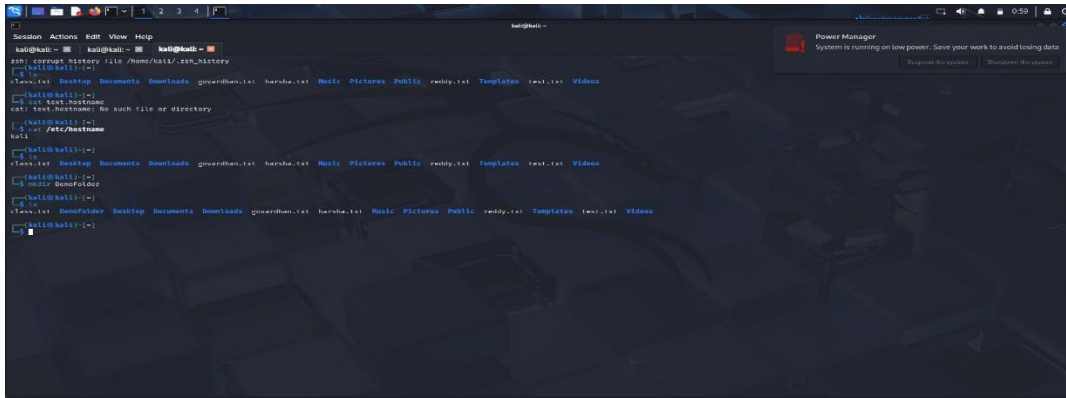
Ls

Cat /etc/hostname

Mkdir DemoFolder

ls

Output:



```
kali@kali:~$ cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
games:x:4:65534:games:/usr/games:/usr/sbin/nologin
ftp:x:5:5:ftp:/var:/usr/sbin/nologin
kali:x:1000:1000:kali:~/kali:/bin/bash

kali@kali:~$ mkdir /tmp/test
kali@kali:~$
```

Result

Successfully executed ls, cat, and mkdir commands.

Experiment 5

System Information Commands

Aim

To execute **uptime**, **date**, and **mv** commands.

Procedure

1. Check system uptime.
2. Display current date.
3. Rename a file using mv.

Commands:

Uptime

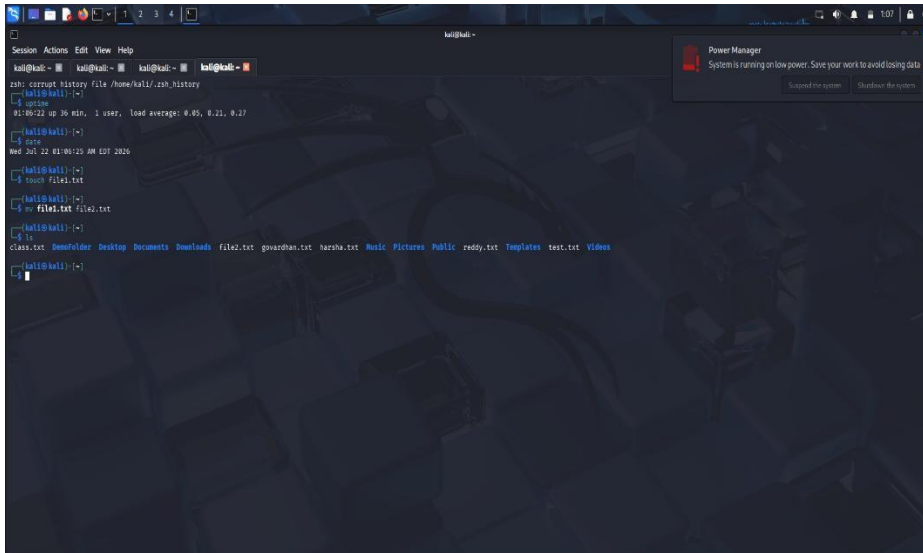
Date

Touch file1.txt

Mv file1.txt file2.txt

Ls

Output:



```
Session Actions Edit View Help
kali@kali ~$ cat /etc/passwd
kali@kali ~$ uptime
01:00:22 up 36 min, 1 user, load average: 0.05, 0.21, 0.27
kali@kali ~$ date
Wed Jul 22 01:00:22 AM EDT 2020
kali@kali ~$ mv file1.txt file2.txt
kali@kali ~$ ls
Class.txt Desktop Documents Downloads file2.txt govardhan.txt harsha.txt Music Pictures Public zeddy.txt Templates test.txt Videos
kali@kali ~$
```

Result

Successfully executed uptime, date, and mv commands.

Experiment 6

File Creation and Download Commands

Aim

To execute touch, pwd, and wget commands.

Theory **touch Command**

Creates an empty file.

touch sample.txt

pwd Command

Displays current working directory.

pwd **wget**

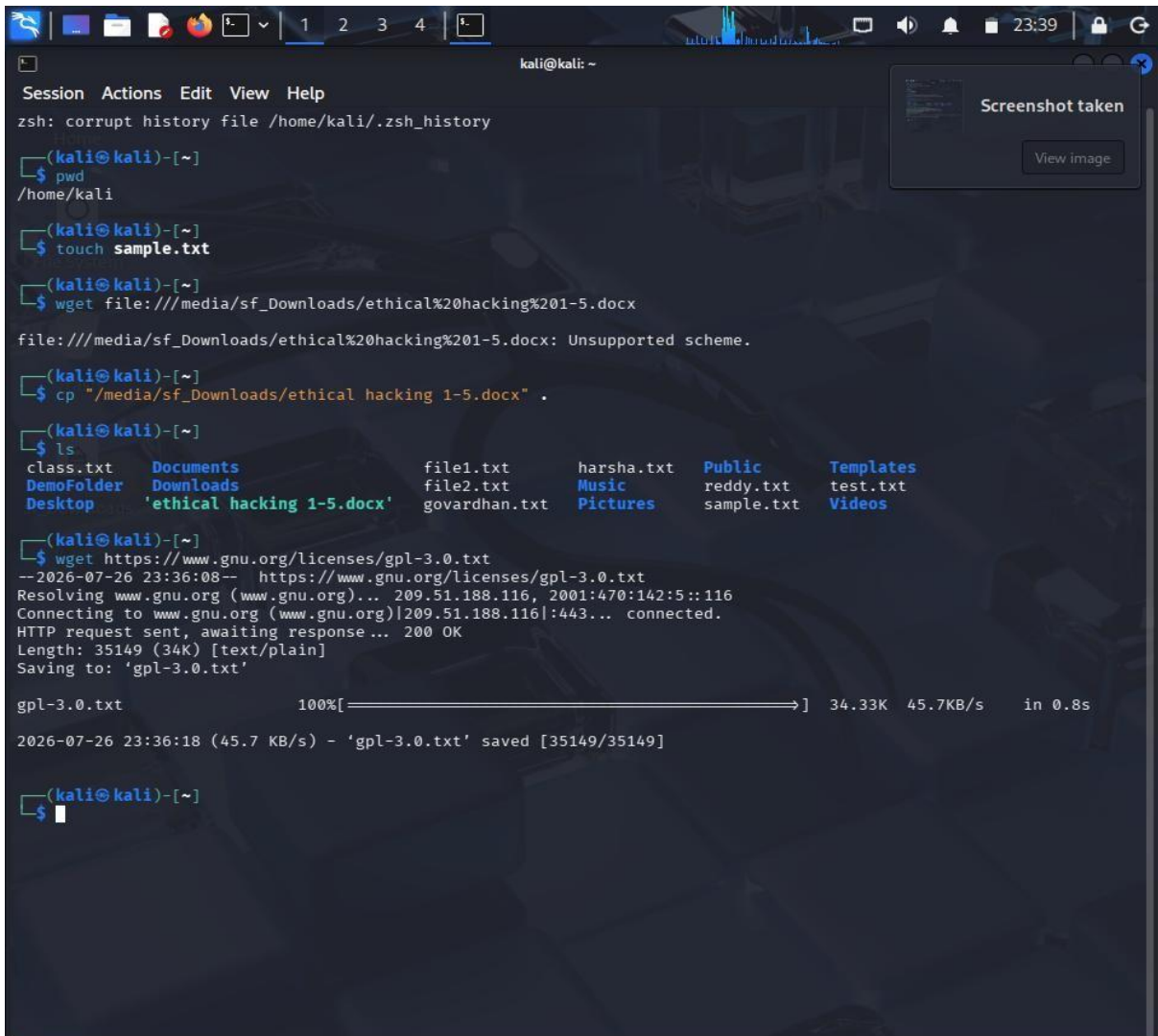
Command

Downloads files from the internet. wget

<https://example.com/file.zip>

Procedure

1. Create a file using touch.
2. Display current directory using pwd.
3. Download a sample file using wget.



```
kali@kali: ~  
Session Actions Edit View Help  
zsh: corrupt history file /home/kali/.zsh_history  
(kali@kali)-[~]  
$ pwd  
/home/kali  
(kali@kali)-[~]  
$ touch sample.txt  
(kali@kali)-[~]  
$ wget file:///media/sf_Downloads/ethical%20hacking%201-5.docx  
file:///media/sf_Downloads/ethical%20hacking%201-5.docx: Unsupported scheme.  
(kali@kali)-[~]  
$ cp "/media/sf_Downloads/ethical hacking 1-5.docx" .  
(kali@kali)-[~]  
$ ls  
class.txt      Documents      file1.txt      harsha.txt     Public         Templates  
DemoFolder    Downloads      file2.txt      Music          reddy.txt      test.txt  
Desktop       'ethical hacking 1-5.docx'  govardhan.txt  Pictures       sample.txt     Videos  
(kali@kali)-[~]  
$ wget https://www.gnu.org/licenses/gpl-3.0.txt  
--2026-07-26 23:36:08-- https://www.gnu.org/licenses/gpl-3.0.txt  
Resolving www.gnu.org (www.gnu.org)... 209.51.188.116, 2001:470:142:5::116  
Connecting to www.gnu.org (www.gnu.org)|209.51.188.116|:443... connected.  
HTTP request sent, awaiting response... 200 OK  
Length: 35149 (34K) [text/plain]  
Saving to: 'gpl-3.0.txt'  
gpl-3.0.txt      100%[=====>] 34.33K  45.7KB/s  in 0.8s  
2026-07-26 23:36:18 (45.7 KB/s) - 'gpl-3.0.txt' saved [35149/35149]  
(kali@kali)-[~]  
$
```

Result

Successfully executed touch, pwd, and wget commands.

Experiment 7

Navigation and Copy Commands

Aim

To execute cd, cp, and ls commands.

Theory **cd Command**

Changes the current directory.

cd Documents **cp**

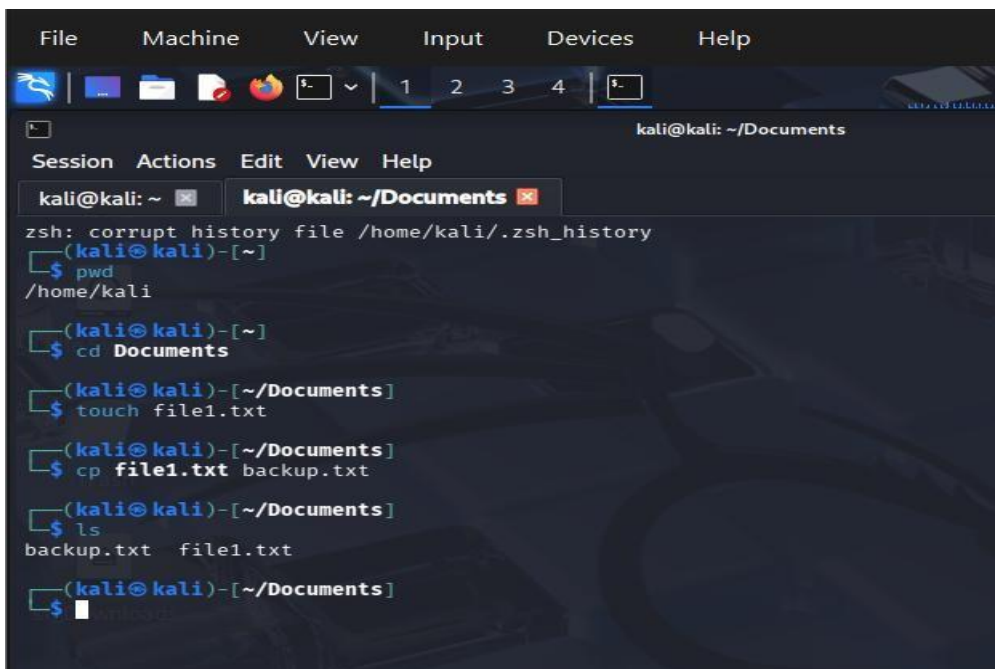
Command

Copies files and directories.

cp file1.txt backup.txt

Procedure

1. Navigate between directories using cd.
2. Copy files using cp.
3. Verify copied files using ls.



```
File Machine View Input Devices Help
kali@kali: ~/Documents
Session Actions Edit View Help
kali@kali: ~ x kali@kali: ~/Documents x
zsh: corrupt history file /home/kali/.zsh_history
(kali@kali)-[~]
$ pwd
/home/kali
(kali@kali)-[~]
$ cd Documents
(kali@kali)-[~/Documents]
$ touch file1.txt
(kali@kali)-[~/Documents]
$ cp file1.txt backup.txt
(kali@kali)-[~/Documents]
$ ls
backup.txt  file1.txt
(kali@kali)-[~/Documents]
$
```

Result

Successfully executed cd, cp, and ls commands.

Experiment 8

Nmap Service Version and OS Detection

Aim

To perform service version detection and operating system identification using Nmap.

Theory

Nmap (Network Mapper) is an open-source network scanning tool used for network discovery and security auditing.

Commands

i) **Detect Service Versions** nmap

```
-sV 192.168.1.10
```

Description:

Determines the versions of services running on target ports.

ii) **Aggressive Scan** nmap -A

```
192.168.1.10
```

Description:

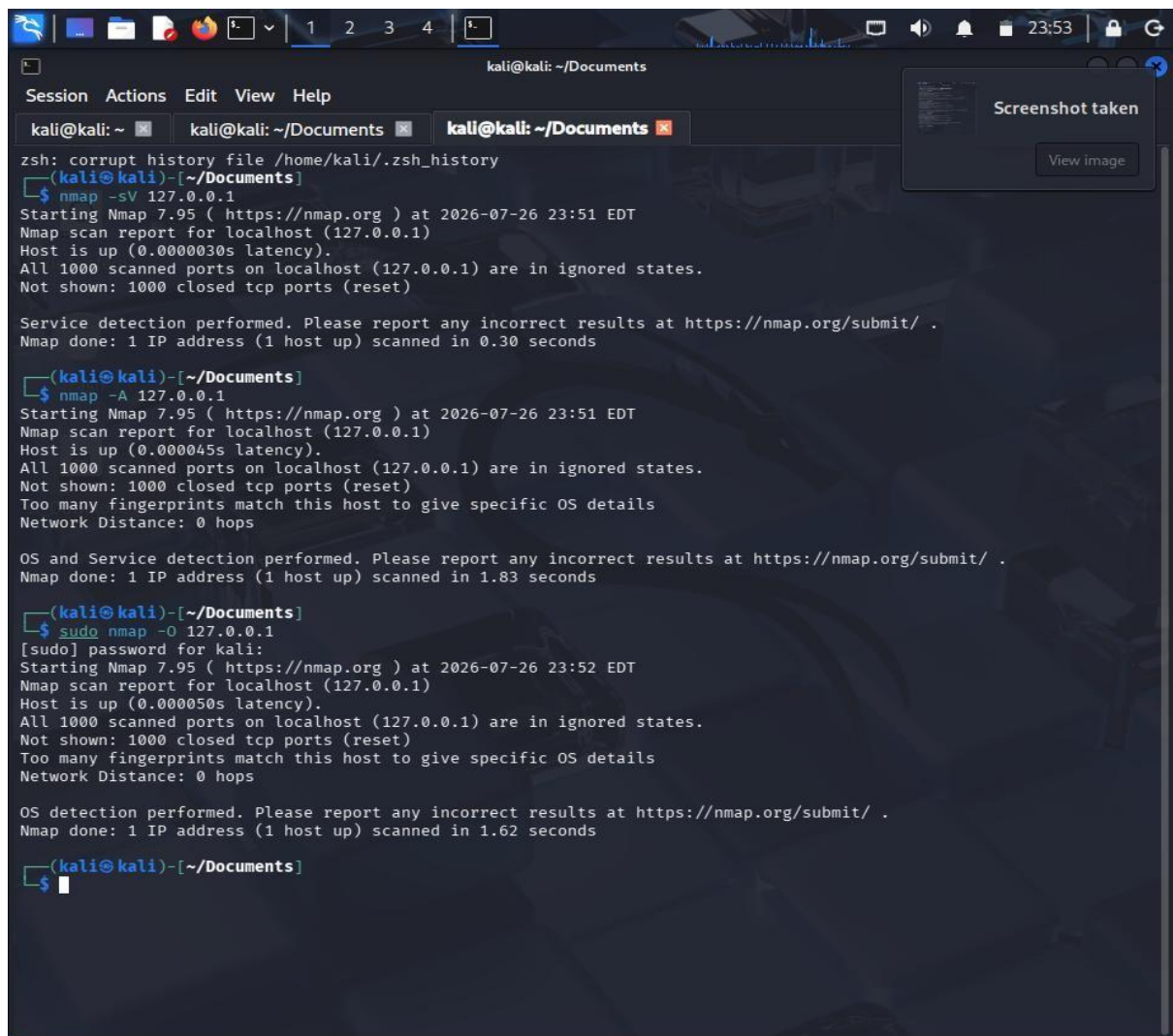
Performs OS detection, version detection, script scanning, and traceroute.

iii) **Operating System Detection**

```
sudo nmap -O 192.168.1.10
```

Description:

Identifies the operating system running on the target host.



```
zsh: corrupt history file /home/kali/.zsh_history
(kali@kali)-[~/Documents]
$ nmap -sV 127.0.0.1
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-26 23:51 EDT
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0000030s latency).
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.
Not shown: 1000 closed tcp ports (reset)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 0.30 seconds

(kali@kali)-[~/Documents]
$ nmap -A 127.0.0.1
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-26 23:51 EDT
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000045s latency).
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.
Not shown: 1000 closed tcp ports (reset)
Too many fingerprints match this host to give specific OS details
Network Distance: 0 hops

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 1.83 seconds

(kali@kali)-[~/Documents]
$ sudo nmap -O 127.0.0.1
[sudo] password for kali:
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-26 23:52 EDT
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000050s latency).
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.
Not shown: 1000 closed tcp ports (reset)
Too many fingerprints match this host to give specific OS details
Network Distance: 0 hops

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 1.62 seconds

(kali@kali)-[~/Documents]
$
```

Result

Successfully performed service version detection and OS identification using Nmap.

Experiment 9

Nmap Timing and Performance Commands

Aim

To perform IDS evasion and timing scans using Nmap.

Theory

Timing options control the speed and stealthiness of Nmap scans.

Commands

i) Polite IDS

Evasion nmap -T2

192.168.1.10

Description:

Slows down scanning to reduce the chance of detection.

ii) Normal Scan

nmap -T3 192.168.1.10

Description:

Default balanced scanning speed.

Procedure

1. Identify target IP address.
2. Execute Nmap timing commands.
3. Observe scan duration and results.
4. Compare scan speeds.

```
zsh: corrupt history file /home/kali/.zsh_history
(kali@kali)-[~/Documents]
$ nmap -T2 127.0.0.1
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-26 23:55 EDT
Stats: 0:00:18 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 4.50% done; ETC: 00:02 (0:06:22 remaining)
Stats: 0:00:23 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 5.70% done; ETC: 00:02 (0:06:21 remaining)
Stats: 0:00:24 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 5.90% done; ETC: 00:02 (0:06:23 remaining)
Stats: 0:01:16 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 18.90% done; ETC: 00:02 (0:05:26 remaining)
Stats: 0:03:59 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 59.60% done; ETC: 00:02 (0:02:43 remaining)
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000099s latency).
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.
Not shown: 1000 closed tcp ports (reset)

Nmap done: 1 IP address (1 host up) scanned in 404.37 seconds

(kali@kali)-[~/Documents]
$ nmap -T3 127.0.0.1
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-27 00:06 EDT
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0000040s latency).
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.
Not shown: 1000 closed tcp ports (reset)

Nmap done: 1 IP address (1 host up) scanned in 0.13 seconds

(kali@kali)-[~/Documents]
$
```

Result

Successfully executed Nmap timing and performance scans in Kali Linux.

Experiment 10: Free, Sort and History Commands

Aim:

To execute and understand the usage of Free, Sort, and History commands in Kali Linux.

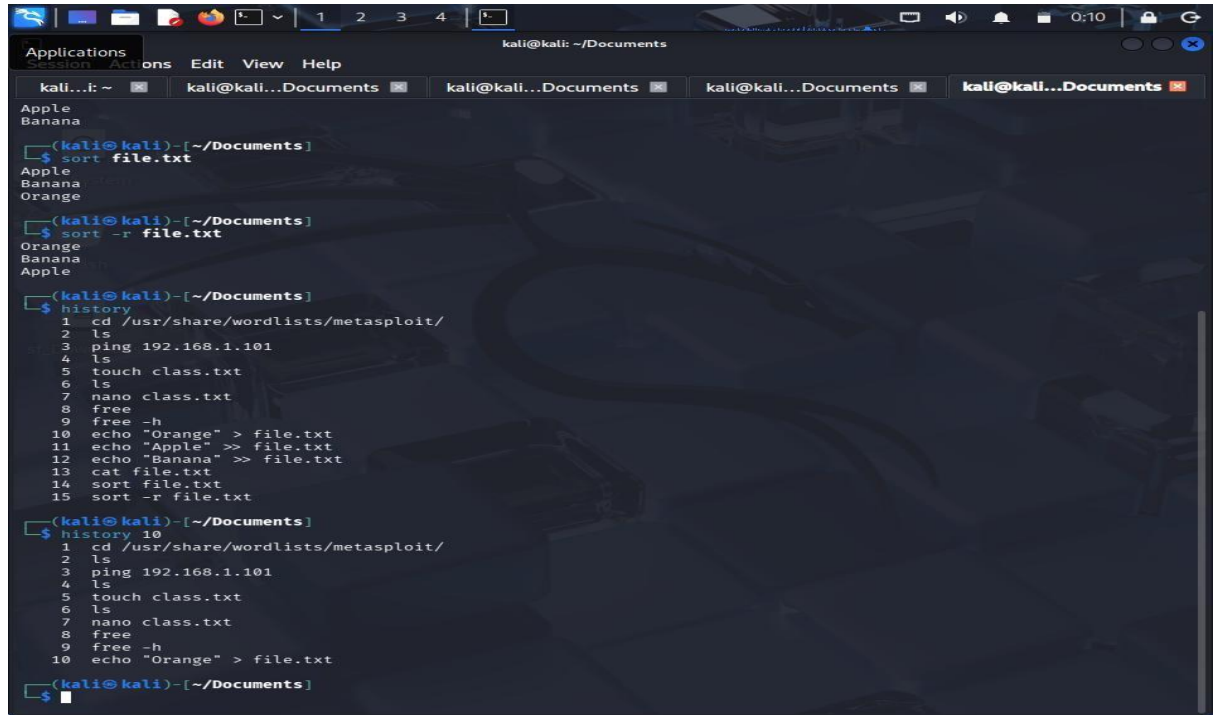
Commands: free

```
free -h sort
file.txt sort
-r file.txt
history
history
10
```

Procedure:

1. Open Terminal in Kali Linux.

2. Execute the `free` command to view memory usage.
3. Create a text file and use the `sort` command.
4. Execute the `history` command to display previously executed commands.



The screenshot shows a Kali Linux terminal window with the following content:

```
kali@kali: ~/Documents
Apple
Banana
(kali@kali)-[~/Documents]
$ sort file.txt
Apple
Banana
Orange
(kali@kali)-[~/Documents]
$ sort -r file.txt
Orange
Banana
Apple
(kali@kali)-[~/Documents]
$ history
1 cd /usr/share/wordlists/metasploit/
2 ls
3 ping 192.168.1.101
4 ls
5 touch class.txt
6 ls
7 nano class.txt
8 free
9 free -h
10 echo "Orange" > file.txt
11 echo "Apple" >> file.txt
12 echo "Banana" >> file.txt
13 cat file.txt
14 sort file.txt
15 sort -r file.txt
(kali@kali)-[~/Documents]
$ history 10
1 cd /usr/share/wordlists/metasploit/
2 ls
3 ping 192.168.1.101
4 ls
5 touch class.txt
6 ls
7 nano class.txt
8 free
9 free -h
10 echo "Orange" > file.txt
(kali@kali)-[~/Documents]
$
```

Result:

Successfully executed Free, Sort, and History commands in Kali Linux.
